

Homework 3

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Problem 1 Let X represent the difference between the number of heads and the number of tails obtained when a coin is tossed n times.

- What are the possible values of X ?
- If the coin is assumed fair, for $n = 3$, what are the probabilities associated with the values that X can take on?
- Using results from part b), write down the probability mass function of X for $n = 3$.
- Following part c), write down the cumulative distribution function of X for $n = 3$.

Solution:

a) Let N_H denote the number of heads appeared in n toss and let N_T denote the number of tails. Since $N_H + N_T = n$, the (N_H, N_T) pair can only have the form

$$(0, n), (1, n-1), (2, n-2), \dots, (n-2, 2), (n-1, 1), (n, 0). \quad (1)$$

Thus $X = N_H - N_T$ can only take the form $n - 2i$ where $0 \leq i \leq n$.

b) Using results from part a), when $n = 3$, the only possible values for X are $3, 1, -1, -3$. Since the coin is fair and $n = 3$ there are $2^3 = 8$ different outcomes with equal probability $1/8$, the resulting probability for different values of X is

- $\{HHH\} \rightarrow X = 3 \rightarrow P(\{HHH\}) = 1/8$
- $\{HHT\}, \{HTH\}, \{THH\} \rightarrow X = 1 \rightarrow P(\{HHT\}) + P(\{HTH\}) + P(\{THH\}) = 3/8$
- $\{TTH\}, \{THT\}, \{HTT\} \rightarrow X = -1 \rightarrow P(\{TTH\}) + P(\{THT\}) + P(\{HTT\}) = 3/8$
- $\{TTT\} \rightarrow X = -3 \rightarrow P(\{TTT\}) = 1/8$

c) Following part b), the probability mass function for X is $p(k) = \mathbb{P}(X = k)$. That is,

$$p(k) = \begin{cases} 1/8, & \text{if } k = 3. \\ 3/8, & \text{if } k = 1. \\ 3/8, & \text{if } k = -1. \\ 1/8, & \text{if } k = -3 \end{cases} \quad (2)$$

and $p(k) = 0$ for other values of k .

d) Following part c), the cumulative distribution function for X is $F(x) = \mathbb{P}(X \leq x)$, and we have

$$F(x) = \begin{cases} 0, & \text{if } x < -3. \\ 1/8, & \text{if } -3 \leq x < -1. \\ 1/2, & \text{if } -1 \leq x < 1. \\ 7/8, & \text{if } 1 \leq x < 3. \\ 1, & \text{if } 3 \leq x \end{cases} \quad (3)$$

Alternative answer: if $X = |N_H - N_T|$ is defined as the absolute value of the difference, then

- $X = |N_H - N_T|$ can take values $n - 2i$ for $0 \leq i \leq \lfloor \frac{n}{2} \rfloor$.
- $P(X = 3) = P(\{HHH\}) + P(\{TTT\}) = 1/4$ and $P(X = 1) = 1 - P(X = 3) = 3/4$.
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$$p(k) = \begin{cases} 1/4, & \text{if } k = 3. \\ 3/4, & \text{if } k = 1 \end{cases} \quad (4)$$

and $p(k) = 0$ for other values of k .

d)

$$F(x) = \begin{cases} 0, & \text{if } x < 1. \\ 3/4, & \text{if } 1 \leq x < 3. \\ 1, & \text{if } 3 \leq x. \end{cases} \quad (5)$$

Problem 2 A product is classified according to the number of defects it contains and the factory that produces it. Let X_1 and X_2 be the random variables that represent the number of defects per unit (taking on possible values of 0, 1, 2, or 3) and the factory number (taking on possible values 1 or 2), respectively. The entries in the table represent the joint possibility mass function of a randomly chosen product.

$X_1 \backslash X_2$	1	2
0	$\frac{1}{8}$	$\frac{1}{16}$
1	$\frac{1}{16}$	$\frac{1}{16}$
2	$\frac{3}{16}$	$\frac{1}{8}$
3	$\frac{1}{8}$	$\frac{1}{4}$

a) Find the marginal probability mass functions of X_1 and X_2 .

b) Find $\mathbb{E}[X_1]$ and $\mathbb{E}[X_2]$.

Solution:

a)

$$\begin{aligned} p_{X_1}(0) &= \mathbb{P}(X_1 = 0) = \mathbb{P}(X_1 = 0, X_2 = 1) + \mathbb{P}(X_1 = 0, X_2 = 2) = 1/8 + 1/16 = 3/16 \\ p_{X_1}(1) &= \mathbb{P}(X_1 = 1) = \mathbb{P}(X_1 = 1, X_2 = 1) + \mathbb{P}(X_1 = 1, X_2 = 2) = 1/16 + 1/16 = 1/8 \\ p_{X_1}(2) &= \mathbb{P}(X_1 = 2) = \mathbb{P}(X_1 = 2, X_2 = 1) + \mathbb{P}(X_1 = 2, X_2 = 2) = 3/16 + 1/8 = 5/16 \\ p_{X_1}(3) &= \mathbb{P}(X_1 = 3) = \mathbb{P}(X_1 = 3, X_2 = 1) + \mathbb{P}(X_1 = 3, X_2 = 2) = 1/8 + 1/4 = 3/8. \end{aligned} \quad (6)$$

and

$$\begin{aligned} p_{X_2}(1) &= \mathbb{P}(X_2 = 1) = \mathbb{P}(X_1 = 0, X_2 = 1) + \mathbb{P}(X_1 = 1, X_2 = 1) + \mathbb{P}(X_1 = 2, X_2 = 1) + \mathbb{P}(X_1 = 3, X_2 = 1) \\ &= 1/8 + 1/16 + 3/16 + 1/8 = 1/2 \\ p_{X_2}(2) &= \mathbb{P}(X_2 = 2) = \mathbb{P}(X_1 = 0, X_2 = 2) + \mathbb{P}(X_1 = 1, X_2 = 2) + \mathbb{P}(X_1 = 2, X_2 = 2) + \mathbb{P}(X_1 = 3, X_2 = 2) \\ &= 1/16 + 1/16 + 1/8 + 1/4 = 1/2. \end{aligned} \quad (7)$$

b)

$$\begin{aligned} \mathbb{E}[X_1] &= 0 \cdot \frac{3}{16} + 1 \cdot \frac{1}{8} + 2 \cdot \frac{5}{16} + 3 \cdot \frac{3}{8} = \frac{15}{8}. \\ \mathbb{E}[X_2] &= 1 \cdot \frac{1}{2} + 2 \cdot \frac{1}{2} = \frac{3}{2}. \end{aligned} \quad (8)$$

Problem 3 A set of five transistors are to be tested, one at a time in a random order, to see which of them are defective. Suppose that three of the five transistors are defective, and let N_1 denote the

number of tests made until the first defective is spotted, and let N_2 denote the number of additional tests until the second defective is spotted. Find the joint probability mass function of N_1 and N_2 .

Solution:

The possible values for N_1 are 1, 2, 3, corresponds to the number of tests need to find the first defective transistor. Meanwhile, we have $N_1 + N_2 \leq 4$, as it takes at most 4 tests to find the first two defective transistors. There are five transistors in total, three of them are defective and two of them are good, then there are $\frac{5!}{3!2!} = 10$ combinations for the positions of good and defective transistors. Given N_1 and N_2 , we know the position of the first two defectives, thus can calculate the corresponding probability.

For example, when $N_1 = 1$ and $N_2 = 1$, the first two transistors are defective, thus there is exact one defective transistor in the last three, and the corresponding number of combination is $\binom{3}{1} = 3$. $\mathbb{P}(N_1 = 1, N_2 = 1) = 3/10 = 0.3$.

Following this idea, we have

$$\begin{aligned} p(1, 1) &= \mathbb{P}(N_1 = 1, N_2 = 1) = 3/10 = 0.3, \\ p(1, 2) &= \mathbb{P}(N_1 = 1, N_2 = 2) = 2/10 = 0.2, \\ p(1, 3) &= \mathbb{P}(N_1 = 1, N_2 = 3) = 1/10 = 0.1, \\ p(2, 1) &= \mathbb{P}(N_1 = 2, N_2 = 1) = 2/10 = 0.2, \\ p(2, 2) &= \mathbb{P}(N_1 = 2, N_2 = 2) = 1/10 = 0.1, \\ p(3, 1) &= \mathbb{P}(N_1 = 3, N_2 = 1) = 1/10 = 0.1. \end{aligned}$$

Alternative solution: If we consider the problem from the perspective of conditional probability, we have:

$$\begin{aligned} p(1, 1) &= P(N_1 = 1)P(N_2 = 1|N_1 = 1) = \frac{3}{5} * \frac{2}{4} = 0.3. \\ p(1, 2) &= P(N_1 = 1)P(N_2 = 2|N_1 = 1) = \frac{3}{5} * (\frac{2}{4} * \frac{2}{3}) = 0.2 \\ p(1, 3) &= P(N_1 = 1)P(N_2 = 3|N_1 = 1) = \frac{3}{5} * (\frac{2}{4} * \frac{1}{3}) = 0.1. \\ p(2, 1) &= P(N_1 = 2)P(N_2 = 1|N_1 = 1) = (\frac{2}{5} * \frac{3}{4}) * \frac{2}{3} = 0.2 \\ p(2, 2) &= P(N_1 = 2)P(N_2 = 2|N_1 = 1) = (\frac{2}{5} * \frac{3}{4}) * \frac{1}{3} = 0.1 \\ p(3, 1) &= P(N_1 = 3)P(N_2 = 1|N_1 = 1) = (\frac{2}{5} * \frac{1}{4} * 1) * 1 = 0.1. \end{aligned}$$

Problem 4 A total number of n children of different heights are placed in a line at random. You select the first child from the line, and walk with her/him along the line, until you encounter a child who is taller, or until you reach the end of the line. If you do encounter a taller child, you also have him/her to accompany you further along the line, until you encounter yet again a taller child or reach the end of the line, etc. Let the random variable X denote the number of children to be selected from the line. What is the expected value of X ?

Solution: Let I_i , $i = 1, 2, \dots, n$. be the random variable indicating whether the i th position on the line incurs a selection (i.e. a taller kid emerges in the i th position)

$$I_i = \begin{cases} 1 & \text{if the } i\text{th position on the line has a selection} \\ 0 & \text{otherwise} \end{cases}$$

We notice that $X = \sum_{i=1}^i I_i$. Then,

$$\begin{aligned} E[I_i] &= P(I_i = 1) = P(\text{the tallest of the first } i \text{ children sits in the } i\text{th position}) = 1/i \\ E[X] &= E\left[\sum_{i=1}^n I_i\right] = \sum_{i=1}^n E[I_i] = \sum_{i=1}^n 1/i. \end{aligned} \tag{9}$$

Problem 5 A community consists of 100 married couples. If 50 members of the community die, what is the expected number of marriages that remain intact? Assume that the set of people who die is equally likely to be any of the $\binom{200}{50}$ groups of size 50. Hint: for $1 \leq i \leq 100$, let

$$X_i = \begin{cases} 1, & \text{if neither member of couple } i \text{ dies.} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

Solution:

$$\mathbb{E}\left[\sum_{i=1}^{100} X_i\right] = \sum_{i=1}^{100} \mathbb{E}[X_i] = \sum_{i=1}^{100} \frac{\binom{198}{50}}{\binom{200}{50}} = \sum_{i=1}^{100} \frac{150 \times 149}{200 \times 199} = 56.156 \tag{11}$$